Eric Chen

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Education

Stanford University Stanford, CA
B.Sc. & M.Sc. Computer Science & B.Sc. Mathematics, **GPA: 4.113/4.0** *Expected: June 2027*
Relevant Coursework: LLMs from Scratch (CS 336), Algorithms, Graduate Probability Theory, Programming Abstractions and Data Structures (C++), Real Analysis, Deep Learning for CV (Python, PyTorch, NumPy), Computer Organization and Systems (C), Statistical ML, Stochastic Processes, NLP with DL, OS Principles (C, C++), Deep RL, Regression & ANOVA, Convex Optimization,

CS/Machine Learning Experiences

Stanford Artificial Intelligence Lab (SAIL): Chain-of-Thought and Multi-Step Pipelines for LLM Judges in Math Stanford, CA
Sole First Author, Machine Learning Researcher *Oct 2025 – Present*

- Designed and ran 13 judge prompting experiments across 5 LLMs using vLLM, batched inference on H200/B200 GPUs.
- Built a 500-problem human-graded benchmark with analytic rubrics; analyzed correlation and variance metrics in SciPy.
- Recruited and led a team of four (one undergrad, Ph.D., postdoc, and professor); **published at ICLR (LLM Reasoning)**.

Stanford Artificial Intelligence Lab (SAIL) and CURIS Summer Research Internship, Stanford University Stanford, CA
First Author, Machine Learning Researcher *Jan 2024 – July 2025*

- Extract and vary data from competitions to create Putnam-AXIOM, 522-large benchmark to evaluate SOTA LLMs' reasoning.
- Evaluated and supervised fine-tuning (FT & PEFT) SOTA LLMs (Mistral-7B, LLAMA-8B, DeepSeek) on a virtual GPU.
- Spearheaded a team of peers; collaborated with Prof. Sanmi Koyejo, **published at ICML, and NeurIPS (MATH-AI track)**.

Deep Reinforcement Learning Final Research Project, Supervised by Prof. Chelsea Finn, Stanford University Stanford, CA
Student Machine Learning Researcher *April 2025 – June 2025*

- Built AlphaZero-style RL agent for Gomoku using MCTS and self-play, boasting 100% win-or-tie rate against human players.
- Awarded **extra credit** by Prof. Finn for novel model architecture, proficiency in PyTorch, GPU optimization, and RL training.

Hudson River Trading (HRT) Summer Internship New York City, NY
Algorithm Developer (Quant Researcher) Intern *May 2026 – Aug 2026*

- Trained equity return prediction models on high-frequency market data across the ingestion, features, and training/validation.
- Analyzed an alternative equity dataset and engineered features from it to improve daily return predictions.

Roblox Corporation Summer Internship San Mateo, CA
Machine Learning & Software Engineer Intern *June 2025 – Sept 2025*

- Designed & productionized transformer and set-transformer-based model to embed & aggregate collections of user data.
- Deployed new model to production at scale w/ Apache Airflow, Kafka, Hive, Spark, AWS S3, and Kubeflow infrastructure.
- **Increased alt-account detection precision 2× to 92%**; enabled enforcement against high priority bad actors.

Publications, Conferences Presentations, and Professional Services

Mixture of Graders: Adaptive Strategy Routing for LLM-Based Mathematical Evaluation
(Lead Author) To Present at ICML 2026 (AI4MATH)).
Putnam-AXIOM: A Functional and Static Benchmark for Measuring Higher Level Mathematical Reasoning
(First Author) Presented at ICML 2025 and NeurIPS 2024 (MATH-AI). Featured on YCombinator HackerNews.
Rethinking LLM Judges: Chain-of-Thought and Multi-Step Pipelines for Math Grading
(Lead Author) To Present at ICLR 2026 (LLM Reasoning).
Using Machine Learning on New Feature Sets Extracted From 3D Models of Broken Animal Bones
Published in the Journal of Human Evolution, Vol. 187 (Feb. 2024). (from AMAAZE, University of Minnesota)
Unitary Conditions for Lamé and Heun Differential Operators
Presented at the 2023 Joint Mathematics Meeting conference. Published in the MIT Math Department. (from MIT PRIMES-USA)
Coarse Graining to Expedite Molecular Dynamics Simulations of Solvated Fibrinogen
Presented at the Materials Research Society, Fall 2023 Meeting. (from the Garcia Research Program, Stony Brook University)
Nominated as Reviewer for the ICML AI4MATH 2025 & NeurIPS MATH-AI 2025 Conference Workshops.

Extracurricular Leadership and Involvement

Association for Computing Machinery (ACM), Machine Learning Lab, Stanford University. *October, 2023 – Present*

- **Won 5th place** as an all-freshman team out of 200+ teams in the 2023 ACM Puzzle Hunt.
- Developed Barcode Buddy, an API-based camera-based allergy and nutrition web app in Treehacks, using Javascript & React.

Business Association of Stanford Entrepreneurial Students (BASES) *September, 2023 – September 2024*

- **One of 30 out of 500+ students (6%)** admitted to BASES Frosh Battalion for exceptional leadership & entrepreneurial skill.

Skills and Interests

- **Programming Languages:** LaTeX (mastery), Python (mastery), C++ (proficient), C (proficient), Java (intermediate).
- **Frameworks and Libraries:** Pytorch, NumPy, Pandas, SciPy, SymPy, Multiprocessing; GitHub, Linux CLI.
- **Languages:** English (fluent), Mandarin (fluent), Spanish (Intermediate).
- **Hobbies**: piano, singing, cycling, escape rooms, competitive video games, poker, exploring the outdoors.